

Total No. of Questions—8]

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[5252]-171

S.E. (Infor. Tech.) (First Semester) EXAMINATION, 2017

DISCRETE STRUCTURE

(2012 COURSE)

Time : Two Hours

Maximum Marks : 50

- N.B. :**
- (i) Answer Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4, Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.
 - (ii) Draw neat diagrams must be drawn wherever necessary.
 - (iii) Figures to the right indicate full marks.
 - (iv) Assume suitable data, if necessary.

1. (a) What are different logical connections ? Define all logical connections with the help of proper example. [6]

(b) P.T. $\frac{1}{4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$ [6]

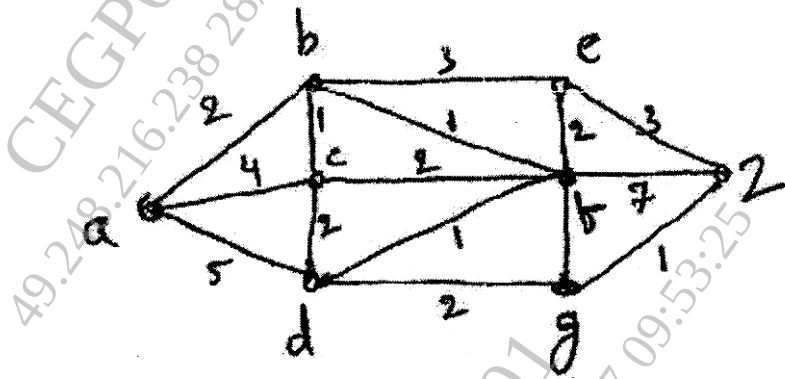
Or

2. (a) A survey of 70 high school students revealed that 35 like folk music, 15 like classical music, and 5 like both. How many of the students surveyed do not like either folk or classical music ? [6]

- (b) What is an equivalent relation ? Identify whether the given relation is an equivalent relation where
 $A = \{1, 2, 3, 4\}$, $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 3), (1, 3), (4, 1), (4, 4)\}$. Also find the equivalent classes. [6]

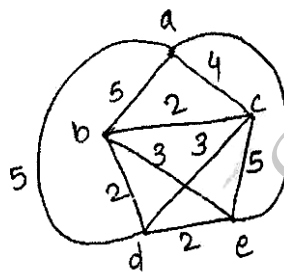
P.T.O.

3. (a) What is monoid ? Show that the algebraic system $(A, +)$ is a monoid, where A is set of integers and $+$ is a binary operation giving addition of two integers. [6]
- (b) Find the shortest path from a to z , using Dijkstra's Algorithm. [6]



Or

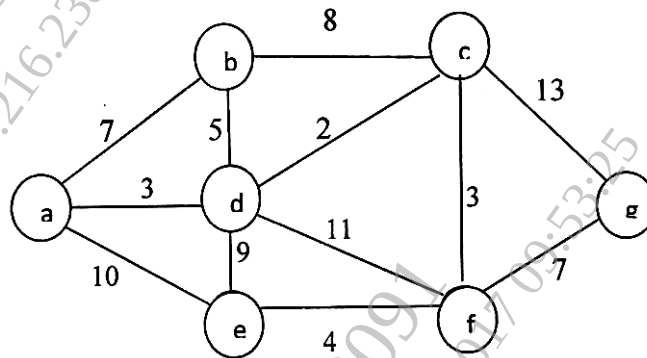
4. (a) Show that the graph G and G^* are isomorphic $G = (V, E)$ and $G^* = (V^*, E^*)$ given by : [6]
- $G = ({a, b, c, d}, \{(a, b), (a, d), (b, d), (c, d), (c, b), (b, c)\})$.
- $G^* = (\{(1, 2, 3, 4)\}, \{(1, 2), (2, 3), (3, 1), (3, 4), (4, 3), (4, 2)\})$.
- (b) (i) Define Graph Coloring. [2]
- (ii) User nearest neighbor method to find Hamiltonian circuit starting from a , find its weight. [4]



5. (a) For the following set of weight, construct the optimal binary prefix tree. For each of the weight in the set, give the corresponding prefix code. [6]

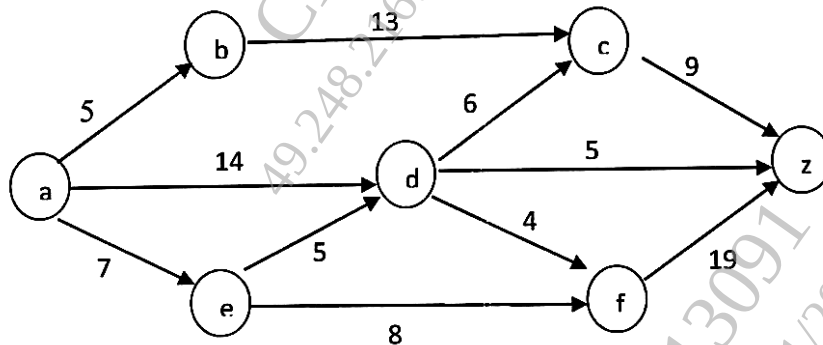
3, 5, 19, 2, 12, 8, 16, 7

- (b) Determine the minimum spanning tree using prims algorithm for the following graph : [6]



Or

6. (a) A find maximum flow in the following transport network. [6]



- (b) Represent the following expression as : [6]

- (1) A binary tree
- (2) Postfix Notation
- (3) Prefix Notation.

$$((a + b) * c) / ((d - e) ^ f)$$

7. (a) How many auto license number plates can be created with 3 alphabets followed by 4 digits if : [4]
(1) Repetition of both alphabets and numbers is allowed.
(2) Repetition of both alphabets and number is not allowed.
- (b) In how many ways can five examinations be scheduled in a week so that no two examinations scheduled on same day considering Sunday as holiday ? [3]
- (c) A committee of 5 members is to be formed from a group of 7 men and 6 women. What is the probability that :
(1) At least 3 women are part of the committee.
(2) All committee members are either men or women. [6]

Or

8. (a) Out of 15 employees in a software company, a group of 5 employees is to be sent for for 'Linux Administration and Networking' training of one month. [7]
(1) In how many ways can the 5 employees be selected ?
(2) What if there are 2 employees who refuse to go together for training ?
(3) What if there are 2 employees who want to go together i.e. either they go together or they do not go for training ?
- (b) Two cards are drawn at random from an ordinary deck of well shuffled 52 cards. Find the probability that : [6]
(1) First cards drawn is ace and second cards drawn is face card of spade.
(2) Both are spades.